

Arizona State University Microeconomic Theory Qualifier

Module 5: Screening, Adverse Selection, Insurance, and Credit

Exhaustive but exam-facing lecture-note packet

Purpose

This packet converts the relevant ASU micro qualifier pattern into a problem-solving system. The center of the module is hidden information: write the object, write feasible menus and beliefs, impose IC/IR/zero-profit, and verify no profitable deviation or no-mimic condition. Every canonical model below uses the four-layer answer framing

Object $\rightarrow$ Feasibility $\rightarrow$ Optimality/Equilibrium $\rightarrow$ Verification.

Skill emphasis: formulation/setup, definitions, equilibrium theorems, graphs, computations, proofs
Study block: two 3-hour blocks plus one 2-hour past-qual drill
Output: source inventory, detailed blueprint, full notes, templates, exercises, answer sketches

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1 Orientation and first-page protocol

Pair with Visual Intuition Curriculum: Draw a menu as points in contract space; the exam answer is a path from primitives to constraints to a verified menu.

A screening problem is never just a maximization problem. It is a maximization problem plus a self-selection argument. Your first page should therefore contain four objects before any algebra:

1. the type space and probabilities;
2. the contract space and outside option;
3. the utility/payoff of each type from each contract;
4. the IC, IR, and zero-profit or profit-maximization restrictions.

Reusable template

Default hidden-information answer frame.

Object.

State who moves, what is privately known, what a contract is, and what equilibrium or optimum is requested.

Feasibility.

Write all menu constraints: contract bounds, IC, IR, zero-profit, Bayes consistency if beliefs matter, and participation.

Optimality/Equilibrium.

Reduce the system using binding constraints, single crossing, zero profit, or best-response conditions; solve the remaining equations.

Verification.

Check every dropped constraint, monotonicity, no profitable deviation, no-mimic inequalities, feasibility bounds, and corner cases.

2 Source inventory

Pair with Visual Intuition Curriculum: Place sources on a timeline from “exam prompts” to “lecture models” to “supplementary proof machinery.”

Source	Type	Topics used	Mathematical tools	Extraction notes
MICRO_MANIFEST.md	Project manifest	Module 5 scope, source hierarchy, required packet elements, notation conventions	Four-layer answer; IC/IR; zero profit; monotone menus	Fully readable.
SKILL.md	Skill instructions	Inventory, pattern analysis, blueprint, packet structure, exercises, TeX rules	Packet workflow; quality audit	Fully readable.
GitHub Games	Bayesian Lecture/source folder	Adverse selection, signaling-screening contrast, Rothschild–Stiglitz insurance	Beliefs, zero-profit lines, separating/pooling equilibrium	Directory accessible; local copies of main .tex files used.

Source	Type	Topics used	Mathematical tools	Extraction notes
GitHub Mechanism Design	Lecture/source folder	One-dimensional IC, monotonicity, envelope, revelation principle, direct mechanisms	BIC/DSIC characterization; monotone allocation; envelope	Directory accessible; local copies of main .tex files used.
die026-advsel.tex	Lecture note	Akerlof lemons, adverse selection definitions, market failure, signaling/screening response	Participation cutoff, conditional expectation, constrained efficiency	Fully readable; original note includes figures not embedded here.
die027-spence.tex	Lecture note	Signaling contrast, zero-profit wage logic, least-cost separation bridge to Module 4	WPBE logic, no-mimic constraints	Used only for contrast; no long duplication.
die028-rs.tex	Lecture note	Rothschild–Stiglitz insurance problem, high-risk full insurance, low-risk partial insurance, pooling deviation	State-contingent wealth diagram; zero-profit lines; IC	Fully readable; original note uses image figures, replaced by described diagrams.
die062-icc1d.tex	Mechanism note	Direct mechanisms, BIC/DSIC, expected allocation and transfer, convex utility/envelope	IC characterization: monotone allocation and envelope	Fully readable; used for proof templates.
die064-appns-1d.tex	Problem-set/lecture note	Applications of one-dimensional IC, revenue equivalence, monotone strategies	Envelope, monotonicity verification	Supplementary; mostly Module 6/auctions.
die067-nreg.tex	Mechanism note	Nonregular mechanisms, monotonicity failures, ironing intuition	Regularity, monotone allocation repair	Supplementary; used for trap about monotonicity.
die072-ms.tex	Mechanism note	Bilateral trade and Myerson–Satterthwaite intuition	BIC, IR, no outside subsidy, efficiency impossibility	Supplementary; used for adverse-selection efficiency warning.
module4_...tex	Prior module	Boundary with signaling, disclosure, beliefs	WPBE and separating/pooling language	Read to avoid duplication.
Micro 13A.pdf	Past qual, August 2013	Monopoly insurance with loading factor; risk aversion comparative statics; no-trade threshold	Participation constraint, risk aversion order, profit cutoff	Scanned image; no parsed text; read from rendered image.
Micro 14J.pdf	Past qual, June 2014	Monopoly insurance screening with two risk types; SSCP; monotone contracts; full coverage for high-risk type	IC/IR menu, single crossing, state-contingent graph	Scanned image; no parsed text; read from rendered image.

Source	Type	Topics used	Mathematical tools	Extraction notes
June 19.pdf	Past qual, June 2019	Credit market screening with collateral and two lenders	Zero profit, separating candidate, no-deviation test, monopoly menu setup	Text extracted.
June 20.pdf	Past qual, June 2020	Competitive insurance with continuum of risk types; full-insurance-only equilibrium; two-type screening extension	Cutoff equilibrium, fixed point, zero profit, existence	Text extracted.
June 21.pdf	Past qual, June 2021	Warranty screening with buyer value types; CARA certainty equivalent; supermodularity; binding constraints	Revelation principle, strict supermodularity, top-no-distortion	Text extracted.
June 23.pdf	Past qual, June 2023	Finite-type quasilinear screening; SSCP iff increasing differences; monotone selection; adjacent binding	Single-crossing proof, compact contract set, finite menus	Text extracted.
August 23.pdf	Past qual, August 2023	Multi-type Spence recursion	Single crossing and separating recursion	Cross-module bridge; useful for no-mimic recursion.
August 24.pdf	Past qual, August 2024	Education signaling with pooling/separating thresholds	Indifference curves and separating verification	Cross-module bridge; not central.
03ic-auc-app.pdf	Problem set	Direct mechanism and monotone virtual-value style exercises	IC, monotone strategies, reserve/entry fee logic	Supplementary; not central to Module 5.
Mas-Colell–Whinston–Green, Kreps	Textbook references in manifest	Standard screening and insurance vocabulary	Separating/pooling, IC reduction, RS existence logic	No page excerpts supplied; used as background only.
Maschler–Zamir–Solan game theory book	Supplementary reference requested	Game-form discipline: strategies, histories, equilibrium verification	Sequential rationality, no-deviation logic	No file supplied; used only as high-level game-theory discipline.

3 Detailed blueprint

Pair with Visual Intuition Curriculum: Draw a two-column map: monopoly screening on the left, competitive screening on the right; both pass through IC/IR before reaching outcomes.

3.1 Module objective

Pair with Visual Intuition Curriculum: Show a menu point, then label it by the four questions it must answer: who chooses it, who wants it, who pays for it, and who might deviate.

After this module, you should be able to take any hidden-information insurance, warranty, product-line, credit, or lemons problem and write a complete exam answer without waiting for inspiration. The target deliverables are:

- formulate the contract/menu problem from primitives;
- state IC, IR, zero-profit, and equilibrium definitions precisely;
- prove monotone menus from strict single crossing or increasing differences;
- identify binding constraints and reduce a two-type menu to two or three equations;
- draw zero-profit lines and indifference curves in insurance/credit diagrams;
- compute separating, pooling, no-trade, and cutoff equilibria;
- verify no profitable deviation in competitive screening games.

3.2 Prerequisites

Pair with Visual Intuition Curriculum: Draw arrows from Module 1 single crossing, Module 3 equilibrium verification, and Module 4 beliefs into the screening menu.

You need: expected utility and risk aversion; KKT and constrained maximization; strict single crossing/increasing differences; pure-strategy SPE; direct revelation; and the Module 4 distinction between signaling by the informed side and screening by the uninformed side. Macro intuition is helpful: credit screening is a static version of collateral constraints and external-finance premia; insurance adverse selection is risk-sharing with endogenous market participation; lemons is a participation cutoff that shifts the composition of trade.

3.3 Canonical models and why they are here

Pair with Visual Intuition Curriculum: Make one graph for each canonical model: menu lattice, state-contingent wealth, credit collateral, and lemons fixed point.

Canonical model	Exam task	Core equations	Verification target
Hidden-information menu grammar	Set up any screening problem	IC, IR, profit, participation	No missing type or off-menu outside option
Single-crossing monotonicity	Prove higher type receives weakly higher quantity/-coverage/warranty	Add two IC inequalities; use increasing differences	Monotone allocation and transfer
Two-type monopoly screening	Reduce to binding low IR and high IC	$T_L = v_L(q_L)$; $T_H = v_H(q_H) - v_H(q_L) + v_L(q_L)$	Low IC and high IR slack/valid
Warranty screening	Solve ASU June 2021 Q3 style menu	CARA certainty equivalent; $v_{\theta w} > 0$	Higher p, w for high type; low PC and high IC bind
Monopoly insurance screening	Solve ASU June 2014 Q3 style menu	$U_i(x_i, t_i)$; $t_i - p_i x_i$; high full coverage	Top type full insurance; monotone contracts

Canonical model	Exam task	Core equations	Verification target
RS competitive insurance	Find separating candidate and deviation test	$t_i = p_i x_i$; $U_H(C_H) = U_H(C_L)$	No profitable cream-skimming/pooling deviation
Full-insurance adverse selection	ASU June 2020 Q2 continuum	cutoff $\tau(\theta) = t$; zero profit $t = \ell E[\theta \mid \theta \geq \theta^*]$	Existence/no-trade/trade fixed point
Credit screening with collateral	ASU June 2019 Q4	zero profit in (r, x) ; high no collateral; low collateral	Candidate within $x \in [0, 1]$ and no deviation
Akerlof lemons	Market breakdown and regulation cutoff	$B(p) = E[v(x) \mid x \leq p]$; trade if $B(p) \geq p$	Composition-consistent price/cutoff
Finite-type adjacent binding	ASU June 2023 Q4 proof style	adjacent IC/IR; recursive transfers	All adjacent and global IC

3.4 Study plan

Pair with Visual Intuition Curriculum: Draw the study path as a sequence: constraints, monotonicity proof, monopoly reduction, competitive deviation test, old-qual timed write-up.

1. **Block 1, 3 hours:** Sections 7–9. Drill IC/IR setup and the single-crossing proof until the inequalities are automatic.
2. **Block 2, 3 hours:** Sections 10–14. Draw the insurance and credit diagrams, compute candidate menus, and practice no-deviation checks.
3. **Block 3, 2 hours:** Work old-qual prompts: 2014 Q3, 2019 Q4, 2020 Q2, 2021 Q3, 2023 Q4. Spend the first 10 minutes only on setup and the last 10 minutes only on verification.

4 Module map

Pair with Visual Intuition Curriculum: Use one row per archetype; the “trap” column is what you check in the last five minutes of an exam answer.

Subtopic	Why it matters	Exam appearances / prototype	Core technique	Common trap
IC/IR template	Every screening problem begins here	2014 Q3, 2021 Q3, 2023 Q4	Write all pairwise IC and IR before reducing	Dropping an IC before proving single crossing
Single crossing	Turns many IC constraints into monotonicity and adjacent constraints	2023 Q4, 2014 Q3, signaling bridges	Add IC inequalities; use increasing differences	Reversing which type values quality more
Binding constraints	Converts menu design into equations	2021 Q3, 2023 Q4	Low IR binds; high IC binds; top no distortion	Binding the high IR in standard monopoly screening

Subtopic	Why it matters	Exam appearances / prototype	Core technique	Common trap
Warranty screening	Combines risk and product quality	2021 Q3	CARA certainty equivalent; supermodularity	Forgetting warranty changes risk, not just mean payoff
Monopoly insurance	Classic private risk problem	2014 Q3, 2013 Q1	State-contingent wealth; full coverage for top risk	Claiming both types get full insurance under private info
Competitive insurance	Separating candidate plus entry/deviation test	2020 Q2, RS notes	Zero-profit lines; no profitable deviations	Stopping at candidate without deviation check
Full-insurance market	Equilibrium can be no trade, trade, or multiple fixed points	2020 Q2	WTP cutoff and average-cost fixed point	Pricing by unconditional average risk
Credit screening	Collateral is inefficient but screens low-risk borrowers	2019 Q4	Zero-profit in (r, x) ; high-risk IC binds	Wrong sign on collateral or forgetting feasibility $x \leq 1$
Lemons	Adverse selection as composition feedback	2015 spring, advsel notes	Conditional expectation fixed point	Treating buyer expectation as unconditional
Finite-type menus	Qual proof archetype for monotone selection	2023 Q4	Adjacent IC and recursive transfers	Proving local IC but not global IC

5 Notation and definitions

Pair with Visual Intuition Curriculum: On one page, draw type on the horizontal axis and contract quality on the vertical axis; IC forces the graph upward.

Types are indexed by $i \in \{L, H\}$ unless otherwise stated. In insurance, $p_H > p_L$ means the high type is a high-risk type. In credit, $q_H > q_L$ means the high type is a high default-risk borrower. In product or warranty screening, $\theta_H > \theta_L$ means the high type has higher willingness to pay.

A contract is denoted $c = (q, T)$ in abstract quasilinear screening, (x, t) in insurance, (p, w) in warranty problems, and (r, x) in credit. A menu is $(c_i)_{i \in \Theta}$.

Definition / theorem

IC and IR. For utility $U_i(c)$ and outside utility $\bar{U}_i$, a menu is incentive compatible and individually rational if

$$\begin{aligned} \text{IC}_i : & \quad U_i(c_i) \geq U_i(c_j) \quad \text{for all } j, \\ \text{IR}_i : & \quad U_i(c_i) \geq \bar{U}_i. \end{aligned}$$

In a direct mechanism, truthful reporting is optimal exactly when the induced menu satisfies these inequalities.

Definition / theorem

Strict single crossing / increasing differences. For quasilinear utility $u(q, T, \theta) = f(q, \theta) - T$, strict increasing differences in (q, θ) means that for $q' > q$ and $\theta' > \theta$,

$$f(q', \theta') - f(q, \theta') > f(q', \theta) - f(q, \theta).$$

Equivalently, higher types have a higher incremental willingness to pay for higher q .

Definition / theorem

Competitive screening equilibrium. Firms simultaneously offer finite menus. Each consumer/borrower type chooses a utility-maximizing offered contract or the outside option. An equilibrium menu must satisfy consumer optimality, firm optimality, zero profit on contracts actually sold in a competitive equilibrium, and no profitable finite-menu deviation by any firm. The last clause is the one most often missed.

6 Pattern recognition and exam decision tree

Pair with Visual Intuition Curriculum: Draw a flowchart: private type? uninformed principal? competitive firms? then choose monopoly IC reduction or competitive zero-profit deviation test.

1. **If the problem says “monopolist/principal offers a menu,”** write the full IC/IR problem and then try to prove monotonicity and binding constraints.
2. **If the problem says “identical firms compete by offering contracts,”** do not solve a single maximization problem first. Draw zero-profit lines, derive the separating candidate, then test deviations.
3. **If only one contract is allowed,** adverse selection becomes a participation/cutoff fixed point: who buys at price t , and does t cover average cost among buyers?
4. **If quality or coverage is chosen by type,** check strict single crossing before claiming higher type gets higher quality.
5. **If collateral appears,** expect an inefficient instrument: it destroys surplus but sorts types.

7 Canonical Model 1: Hidden-information menu grammar

Pair with Visual Intuition Curriculum: Draw each contract as a labeled point; each type picks the point on the highest feasible indifference curve.

Object.

An uninformed principal or set of firms offers contracts to privately informed agents. The object is a menu $\{c_i\}_{i \in \Theta}$ and an assignment rule saying type i selects c_i . In a monopoly problem, the object maximizes expected profit. In a competitive problem, the object is a menu that survives entry and earns zero profit on active contracts.

Feasibility.

A feasible direct menu satisfies contract bounds, participation, and self-selection:

$$\begin{aligned} U_i(c_i) &\geq \bar{U}_i && \text{for each } i, \\ U_i(c_i) &\geq U_i(c_j) && \text{for each } i, j. \end{aligned}$$

If firms compete, also record the profit of each active contract and the no-deviation condition.

Optimality/Equilibrium.

In monopoly screening, solve

$$\max_{(c_i)} \sum_i \pi_i \Pi(c_i, i) \quad \text{s.t. IC and IR.}$$

In competitive screening, candidate contracts usually lie on type-specific zero-profit lines; separating candidates are pinned down by a zero-profit equation plus a binding no-mimic equation.

Verification.

Verify all IC and IR constraints, not only the ones used. For competitive problems, verify no firm can profitably offer a contract or menu that attracts one type or both types. For direct mechanisms, state why revelation principle or self-selection justifies focusing on menus.

Exam-facing takeaway

A grader rewards a clean setup even if the algebra later gets messy. In this module, the setup is usually half the problem.

Proof move: revelation-principle reduction

Pair with Visual Intuition Curriculum: Draw an indirect mechanism as a black box and replace it by a direct menu with the same chosen outcomes.

If an indirect mechanism has an equilibrium in which type i ends with contract c_i , define the direct mechanism that asks the agent to report a type and assigns c_i after report i . The original equilibrium incentives imply $U_i(c_i) \geq U_i(c_j)$ for all j , so truthful reporting is incentive compatible. This is why ASU prompts that ask to “use the revelation principle” expect the direct IC/IR constraints, not a description of all possible messages.

8 Canonical Model 2: Single crossing and monotone menus

Pair with Visual Intuition Curriculum: Draw two indifference curves in (q, T) space; higher types’ curves are flatter when T is vertical and willingness to pay is higher.

Object.

Prove that higher types receive weakly higher contract quality, coverage, warranty, or quantity. For quasilinear contracts $c_i = (q_i, T_i)$, type θ_i has utility $u(q, T, \theta_i) = f(q, \theta_i) - T$.

Feasibility.

The only feasibility needed for the monotonicity proof is IC:

$$\begin{aligned} f(q_i, \theta_i) - T_i &\geq f(q_j, \theta_i) - T_j, \\ f(q_j, \theta_j) - T_j &\geq f(q_i, \theta_j) - T_i. \end{aligned}$$

Assume $\theta_j > \theta_i$ and strict increasing differences in (q, θ) .

Optimality/Equilibrium.

Add the two IC inequalities:

$$f(q_i, \theta_i) - f(q_j, \theta_i) \geq f(q_i, \theta_j) - f(q_j, \theta_j).$$

If $q_i > q_j$, strict increasing differences says the reverse strict inequality must hold. Therefore $q_j \geq q_i$.

Verification.

Once $q_j \geq q_i$, use low-type IC to get

$$T_j - T_i \geq f(q_j, \theta_i) - f(q_i, \theta_i) \geq 0,$$

so transfers are also weakly ordered when f is increasing in q . State whether the problem asks for quantity only or both quantity and transfer monotonicity.

Qualifier trap

Single crossing does not say “the high type has higher utility.” It says the high type has a higher incremental willingness to pay for the sorting dimension. Always identify that dimension before writing inequalities.

9 Canonical Model 3: Two-type monopoly screening and binding constraints

Pair with Visual Intuition Curriculum: Draw two menu points; the low type sits on the outside-option curve and the high type’s indifference curve passes through both contracts.

Let $\Pr(H) = \beta \in (0, 1)$, $\Pr(L) = 1 - \beta$, $\theta_H > \theta_L$, and $u_i(q, T) = v_i(q) - T$ with $v_H - v_L$ strictly increasing in q . The principal’s cost is $c(q)$ and profit from contract (q, T) is $T - c(q)$. Normalize outside utility to zero.

Object.

Find the profit-maximizing menu $(q_L, T_L), (q_H, T_H)$.

Feasibility.

Write all constraints:

$$\begin{array}{ll} \text{IR}_L : & v_L(q_L) - T_L \geq 0, \\ \text{IR}_H : & v_H(q_H) - T_H \geq 0, \\ \text{IC}_L : & v_L(q_L) - T_L \geq v_L(q_H) - T_H, \\ \text{IC}_H : & v_H(q_H) - T_H \geq v_H(q_L) - T_L. \end{array}$$

Monotonicity from Section 8 gives $q_H \geq q_L$.

Optimality/Equilibrium.

In the standard two-type monopoly case, IR_L and IC_H bind. Hence

$$\begin{aligned} T_L &= v_L(q_L), \\ T_H &= v_H(q_H) - [v_H(q_L) - v_L(q_L)]. \end{aligned}$$

Substitute into expected profit:

$$\begin{aligned}\Pi(q_L, q_H) &= (1 - \beta)[v_L(q_L) - c(q_L)] \\ &\quad + \beta[v_H(q_H) - c(q_H) - v_H(q_L) + v_L(q_L)].\end{aligned}$$

If differentiable and interior,

$$\begin{aligned}v'_H(q_H) &= c'(q_H), \\ (1 - \beta)[v'_L(q_L) - c'(q_L)] &= \beta[v'_H(q_L) - v'_L(q_L)].\end{aligned}$$

The high type is efficient at the top; the low type is distorted downward.

Verification.

Check $q_H \geq q_L$, $q_i \geq 0$, any upper bounds, and the dropped constraints. High IR holds because

$$U_H(c_H) = v_H(q_L) - v_L(q_L) \geq 0$$

under the usual normalization and monotonicity. Low IC holds because $q_H \geq q_L$ and increasing differences make the high type's rent from mimicking low at least the low type's rent from mimicking high.

Exam-facing takeaway

The canonical phrase is: “low participation and high incentive compatibility bind; high participation is slack and low IC follows from monotonicity/single crossing.” Then prove it or cite the proof you just wrote.

Why the low type is distorted downward

Pair with Visual Intuition Curriculum: Draw the marginal-surplus curve for the low type and subtract the information-rent wedge.

The principal would set $v'_L(q_L) = c'(q_L)$ under complete information. Under private information, increasing q_L raises the low type's surplus but also raises the high type's information rent $v_H(q_L) - v_L(q_L)$. The low-type first-order condition therefore subtracts the marginal rent wedge $v'_H(q_L) - v'_L(q_L) > 0$.

10 Canonical Model 4: Warranty screening with private valuation

Pair with Visual Intuition Curriculum: Draw warranty w on the horizontal axis and price p on the vertical axis; the high-valuation buyer is willing to pay more for insurance against breakdown.

This is the ASU June 2021 style problem. A product works with probability $q \in (0, 1)$. Type $\theta \in \{\theta_L, \theta_H\}$, with $\theta_H > \theta_L$, values a working product at θ . A contract is price p and warranty/reimbursement w if the product breaks. The seller's expected profit is $p - (1 - q)w$. Buyer expected utility is

$$qu(\theta - p) + (1 - q)u(w - p), \quad u' > 0, \quad u'' < 0.$$

Object.

Set up and solve/characterize the seller's optimal menu $\{(p_L, w_L), (p_H, w_H)\}$.

Feasibility.

By the revelation principle, impose

$$\begin{aligned} qu(\theta_i - p_i) + (1 - q)u(w_i - p_i) &\geq u(0) && \text{IR}_i, \\ qu(\theta_i - p_i) + (1 - q)u(w_i - p_i) &\geq qu(\theta_i - p_j) + (1 - q)u(w_j - p_j) && \text{IC}_{ij}. \end{aligned}$$

With CARA, $u(c) = -e^{-rc}$, the certainty equivalent is

$$\begin{aligned} CE(\theta, p, w) &= -\frac{1}{r} \log \left(qe^{-r(\theta-p)} + (1-q)e^{-r(w-p)} \right) \\ &= v(\theta, w) - p, \end{aligned}$$

where

$$v(\theta, w) = -\frac{1}{r} \log \left(qe^{-r\theta} + (1-q)e^{-rw} \right).$$

Thus the problem becomes quasilinear in price: utility $v(\theta, w) - p$.

Optimality/Equilibrium.

Show v has strict increasing differences. A direct derivative verifies

$$\frac{\partial v}{\partial w} = \frac{(1-q)e^{-rw}}{qe^{-r\theta} + (1-q)e^{-rw}},$$

and this derivative is increasing in θ , so $v_{\theta w} > 0$. Therefore higher types receive weakly higher warranty and price. The standard two-type reduction gives binding low IR and high IC:

$$\begin{aligned} p_L &= v(\theta_L, w_L), \\ p_H &= v(\theta_H, w_H) - v(\theta_H, w_L) + v(\theta_L, w_L). \end{aligned}$$

Substitution gives seller objective

$$(1 - \beta)[v(\theta_L, w_L) - (1 - q)w_L] + \beta[v(\theta_H, w_H) - (1 - q)w_H - v(\theta_H, w_L) + v(\theta_L, w_L)].$$

The high warranty satisfies the no-distortion condition $v_w(\theta_H, w_H) = 1 - q$ when interior; w_L is distorted downward by the rent wedge.

Verification.

Check $w_H \geq w_L$, $p_H \geq p_L$, low IC, high IR, and any warranty bounds. If q is also chosen at cost, do not simply rank q_H and q_L without proving increasing differences in (θ, q) as well as (θ, w) .

Qualifier trap

The warranty problem is not just a mean-valuation problem. Warranty changes the distribution of consumption across working and breakdown states, so the certainty-equivalent step is what makes the problem a clean screening problem.

11 Canonical Model 5: Monopoly insurance screening

Pair with Visual Intuition Curriculum: Draw accident-state wealth on the horizontal axis and no-accident wealth on the vertical axis; full insurance is the 45-degree line.

Types differ by loss probability p_i , with $0 < p_L < p_H < 1$. A contract (x, t) gives coverage $x \in [0, \ell]$ in the loss state and charges premium t in both states. Wealth is $w - \ell + x - t$ if loss occurs and $w - t$ otherwise. Type i 's expected utility is

$$U_i(x, t) = p_i u(w - \ell + x - t) + (1 - p_i) u(w - t),$$

where $u' > 0$ and $u'' < 0$. The insurer's expected profit on type i is $t - p_i x$.

Object.

A monopoly insurer chooses a menu $((x_L, t_L), (x_H, t_H))$ to maximize expected profit subject to IC and participation.

Feasibility.

Write

$$\begin{aligned} U_i(x_i, t_i) &\geq U_i(0, 0) && \text{IR}_i, \\ U_i(x_i, t_i) &\geq U_i(x_j, t_j) && \text{IC}_{ij}, \\ 0 &\leq x_i \leq \ell. \end{aligned}$$

Single crossing holds because high-risk types value coverage more at a fixed premium.

Optimality/Equilibrium.

The standard monopoly screening pattern is: low-risk IR and high-risk IC bind, and the high-risk type receives full insurance, $x_H = \ell$. The binding equations are

$$\begin{aligned} U_L(x_L, t_L) &= U_L(0, 0), \\ U_H(\ell, t_H) &= U_H(x_L, t_L). \end{aligned}$$

The remaining choice of x_L trades off surplus from insuring the low-risk type against information rent left to the high-risk type.

Verification.

Verify $x_H = \ell$ by the top-no-distortion argument. Along a high-risk indifference curve,

$$\left. \frac{dt}{dx} \right|_{U_H} = \frac{p_H u'(w - \ell + x - t)}{p_H u'(w - \ell + x - t) + (1 - p_H) u'(w - t)}.$$

If $x < \ell$, loss-state wealth is below no-loss wealth, so $u'(w - \ell + x - t) > u'(w - t)$ and $dt/dx > p_H$. Increasing x and t along the high-risk indifference curve raises profit $dt - p_H dx > 0$ until full insurance. Then check low IC and high IR.

Exam-facing takeaway

A concise proof that the top risk gets full coverage is often better than calculus on the whole menu. The geometry is: the high-risk indifference curve is steeper than the high-risk zero-profit line until the 45-degree full-insurance line.

Monopoly insurance with a loading factor

Pair with Visual Intuition Curriculum: Draw the insurer's isoprofit line steeper than the actuarially fair line; the loading creates underinsurance even without adverse selection.

In the August 2013 style monopoly-insurance problem, a single known risk type faces insurer cost γpx with $\gamma > 1$. A contract (x, t) solves

$$\max_{0 \leq x \leq \ell, t} t - \gamma px \quad \text{s.t.} \quad U(x, t) \geq U(0, 0).$$

The participation constraint binds. If the solution is interior, the premium increase along the consumer's indifference curve equals marginal cost:

$$\frac{pu'(w - \ell + x - t)}{pu'(w - \ell + x - t) + (1 - p)u'(w - t)} = \gamma p.$$

Since the left side equals p at full insurance and $\gamma p > p$, full insurance is not optimal under a positive loading unless a corner or parameter restriction intervenes. More risk aversion raises willingness to pay for risk smoothing; a no-trade threshold in γ is lower for less risk-averse consumers and higher for more risk-averse consumers, holding primitives fixed.

12 Canonical Model 6: Rothschild–Stiglitz competitive insurance

Pair with Visual Intuition Curriculum: Draw zero-profit lines $t = p_L x$ and $t = p_H x$; the high-risk full-insurance contract and the low-risk partial-insurance contract form a separating pair.

Now identical risk-neutral firms compete by offering insurance contracts. Consumers choose their favorite offered contract or remain uninsured. The firm profit from selling (x, t) to type i is $t - p_i x$.

Object.

Characterize a separating competitive equilibrium candidate and test whether it is an equilibrium.

Feasibility.

In any competitive equilibrium, active contracts cannot earn positive profit because a rival can slightly undercut, and cannot earn negative profit because a firm would drop them. Thus type-specific separating contracts lie on zero-profit lines:

$$t_L = p_L x_L, \quad t_H = p_H x_H.$$

IC and IR must hold.

Optimality/Equilibrium.

The RS separating candidate is

$$\begin{aligned} C_H &= (\ell, p_H \ell), \\ C_L &= (x_L, p_L x_L), \end{aligned}$$

where $x_L \in [0, \ell]$ solves the high-risk no-mimic equation

$$U_H(\ell, p_H \ell) = U_H(x_L, p_L x_L).$$

High risks receive full insurance at their actuarially fair premium; low risks receive partial insurance to deter high-risk mimicry.

Verification.

Check low-risk IC and both IR constraints. Then check no profitable deviation. A deviation that attracts only low risks must satisfy $t > p_L x$ and make low risks strictly prefer it. A deviation that attracts both types must satisfy $t > \bar{p}x$, where $\bar{p}$ is the average loss probability among attracted buyers, and make both types prefer it. If such a contract exists, the separating candidate is not an equilibrium. This is why RS equilibria may fail to exist.

Qualifier trap

A separating candidate is not an equilibrium until you test deviations. The most common losing answer draws the RS pair and stops.

Pooling contracts and cream-skimming

Pair with Visual Intuition Curriculum: Draw a pooling point on the pooled zero-profit line and a nearby contract that only low-risk types prefer.

A pooling contract priced at average cost is vulnerable when a rival can offer slightly less coverage at a lower premium that low-risk types prefer and high-risk types reject. The rival earns positive profit because it attracts a better risk pool. On a diagram, this deviation lies above the low-risk zero-profit line and below the high-risk indifference curve through the pooling allocation.

13 Canonical Model 7: Competitive full-insurance adverse selection with a continuum

Pair with Visual Intuition Curriculum: Draw willingness to pay $\tau(\theta)$ and average cost among buyers; equilibria are intersections.

This is the June 2020 full-insurance-only market. A consumer with loss probability $\theta \in [\theta, \bar{\theta}]$ can buy full insurance against loss ℓ at premium t . Full insurance gives sure wealth $w - t$. No insurance gives expected utility $\theta u(w - \ell) + (1 - \theta)u(w)$.

Object.

Define competitive equilibrium premiums and prove existence or characterize no-trade/trade equilibria.

Feasibility.

Define willingness to pay $\tau(\theta)$ by

$$u(w - \tau(\theta)) = \theta u(w - \ell) + (1 - \theta)u(w).$$

Since u is strictly increasing and concave, τ is continuous and increasing in θ . At premium t , buyers are the set

$$B(t) = \{\theta : \tau(\theta) \geq t\}.$$

Optimality/Equilibrium.

A trade equilibrium premium satisfies zero profit among buyers:

$$t = \ell \mathbb{E}[\theta \mid \theta \in B(t)],$$

with $B(t)$ nonempty. No trade is also an equilibrium if every active full-insurance premium either attracts a pool with negative profit or is rejected.

Verification.

Verify consumer optimality via the cutoff condition and firm optimality via zero profit/no profitable entry. For a proposed premium, never use unconditional average risk unless all types buy.

Closed-form log example from June 2020

Pair with Visual Intuition Curriculum: Draw the two functions $1 - 0.2^\theta$ and $0.4(1 + \theta)$; the interior crossing is the trade equilibrium.

For $u(c) = \log c$, $w = 1$, $\ell = 0.8$, and uniform $\theta \sim U[0, 1]$,

$$\log(1 - \tau(\theta)) = \theta \log(0.2), \quad \tau(\theta) = 1 - 0.2^\theta.$$

If the cutoff type is θ^* , buyers are $[\theta^*, 1]$ and zero profit requires

$$t = 0.8 \frac{1 + \theta^*}{2} = 0.4(1 + \theta^*).$$

The cutoff equation is

$$1 - 0.2^{\theta^*} = 0.4(1 + \theta^*).$$

There is an interior solution approximately $\theta^* \approx 0.7391$, giving $t \approx 0.6957$. The endpoint $\theta^* = 1$ has zero mass of buyers and corresponds to no substantive trade, so the economically relevant outcomes are no trade and the interior trade equilibrium.

14 Canonical Model 8: Credit screening with collateral

Pair with Visual Intuition Curriculum: Draw interest r vertically and collateral x horizontally; low-risk borrowers can credibly choose collateral because they are less likely to lose it.

In the June 2019 credit problem, a borrower has default probability $q \in \{q_L, q_H\}$ with $0 < q_L < q_H < 1/2$. A contract (r, x) specifies interest $r \geq 0$ and collateral $x \in [0, 1]$. Borrower utility is

$$U_q(r, x) = 1 - (1 - q)r - qx.$$

The lender's zero-profit expression consistent with the complete-information benchmark in the exam is

$$\Pi_q(r, x) = (1 - q)r - q + qbx,$$

where $b \in (0, 1)$ is the fraction of collateral value the lender realizes in default. Thus collateral is inefficient: it costs the borrower x in default but yields the lender only bx .

Object.

Derive the unique separating SPE candidate and test existence. Also write the collusive monopoly problem if asked.

Feasibility.

Complete-information zero profit solves

$$(1 - q_i)r_i - q_i + q_i b x_i = 0.$$

Because collateral is inefficient, complete information uses $x_i = 0$ and

$$r_i = \frac{q_i}{1 - q_i}.$$

Under private information, require IC, IR, $x_i \in [0, 1]$, and zero profit on active competitive contracts.

Optimality/Equilibrium.

The high-risk type receives the high-risk zero-collateral contract

$$(r_H, x_H) = \left(\frac{q_H}{1 - q_H}, 0 \right).$$

The low-risk contract lies on the low-risk zero-profit line and makes the high-risk borrower indifferent between mimicking and taking the high-risk contract:

$$r_L = \frac{q_L(1 - bx_L)}{1 - q_L},$$

$$U_{q_H}(r_H, 0) = U_{q_H}(r_L, x_L).$$

Solving gives

$$x_L = \frac{q_H - q_L}{q_H(1 - q_L) - bq_L(1 - q_H)},$$

provided this lies in $[0, 1]$; then r_L follows from zero profit.

Verification.

Check low-risk IC, both IR constraints, $x_L \leq 1$, $r_L \geq 0$, and no profitable deviation. If $x_L > 1$, the separating candidate is infeasible; if a pooling or cream-skimming deviation attracts a profitable pool, a pure-strategy SPE fails to exist. In a monopoly version, do not impose zero profit; instead maximize expected lender profit subject to IC and IR.

Exam-facing takeaway

The collateral is not there because it is efficient. It is there because high-default-risk borrowers dislike collateral more, so collateral can separate types.

15 Canonical Model 9: Akerlof lemons and participation fixed points

Pair with Visual Intuition Curriculum: Draw price on the horizontal axis and buyer conditional value $B(p)$ on the vertical axis; trade requires the curve to lie above the 45-degree line.

Sellers privately know quality x with seller reservation value x . Buyers value a quality- x object at $v(x)$ but observe only whether the object is offered for sale. At price p , sellers with $x \leq p$ sell. The buyer's conditional value is

$$B(p) = \mathbb{E}[v(x) \mid x \leq p] = \frac{\int_x^p v(z)f(z) dz}{\int_x^p f(z) dz}.$$

Object.

Find market prices and qualities traded under adverse selection; compare complete information and incomplete information.

Feasibility.

Trade at price p is composition-consistent only if sellers with $x \leq p$ participate and buyers are willing to pay p for the conditional pool:

$$B(p) \geq p.$$

If legal or regulatory restrictions impose $x \geq r$, replace the conditioning set by $x \in [r, p]$.

Optimality/Equilibrium.

A competitive price with trade often solves $B(p) = p$ with the associated seller cutoff. No trade or only lowest-quality trade occurs when the conditional expectation falls below price for all broader pools.

Verification.

Verify both sides: sellers with $x \leq p$ want to sell and buyers' expectation is conditional on exactly that set. If regulation changes the feasible set, recompute the conditional expectation; do not reuse the old pool.

Connection to macro and finance

Pair with Visual Intuition Curriculum: Draw a capital-flow diagram in which a lender prices the average quality of applicants; high-quality borrowers exit first.

The lemons mechanism is a static version of an adverse-composition spiral. A higher required return or lower buyer belief removes good types from the market, which worsens the pool and can shut down trade. In macro-finance language, prices depend on endogenous applicant composition, not only on technology or preferences.

16 Canonical Model 10: Finite-type adjacent binding and recursive menus

Pair with Visual Intuition Curriculum: Draw a staircase of contracts indexed by type; adjacent no-mimic constraints pin down transfers one step at a time.

The June 2023 finite-type problem uses $\Theta = \{\theta_1 < \dots < \theta_n\}$ and utility $u(q, T, \theta) = f(q, \theta) - T$ with strictly increasing differences. The monopolist offers contracts plus the outside option $(0, 0)$.

Object.

Prove monotonicity and characterize binding adjacent constraints in a finite menu.

Feasibility.

IC means each type prefers its assigned contract to every other assigned contract and to the outside option. Compactness of the contract set gives existence of maximizers when requested.

Optimality/Equilibrium.

Monotonicity follows from the two-IC proof. In many finite-type screening problems, transfers are pinned down recursively by adjacent downward IC or IR constraints:

$$f(q_i, \theta_i) - T_i = f(q_{i-1}, \theta_i) - T_{i-1}, \quad i = 1, \dots, n,$$

with $(q_0, T_0) = (0, 0)$ when the outside option is the previous contract for the lowest type. If a constraint were slack, the principal could raise T_i slightly without violating higher-priority constraints until some adjacent constraint binds.

Verification.

Local adjacent IC plus monotone quantities and increasing differences imply global IC; write the telescoping argument if the prompt asks for proof. Always verify the lowest type's participation and the highest type's rent.

Qualifier trap

“Adjacent IC binds” is not a magic phrase. You must say which adjacent constraint binds and why raising the transfer would not break the other constraints until that one binds.

17 Core derivation templates

Pair with Visual Intuition Curriculum: Put these templates on a one-page cheat sheet and practice filling them with primitives from old quals.

17.1 Template A: Two-type monopoly screening

Pair with Visual Intuition Curriculum: Draw low IR and high IC as the two thick constraints; all other constraints are thin checks.

Reusable template

1. Write $U_i(c_j)$ for every type-contract pair.
2. Write four constraints: IR_L , IR_H , IC_L , IC_H .
3. Prove monotonicity using single crossing.
4. Assert and justify binding IR_L and IC_H .
5. Substitute transfers into expected profit.
6. Solve FOCs/corners.
7. Verify dropped constraints and bounds.

17.2 Template B: RS competitive insurance

Pair with Visual Intuition Curriculum: Draw high full insurance first, then slide low coverage down the low-risk zero-profit line until high IC binds.

Reusable template

1. Draw $t = p_L x$, $t = p_H x$, and full coverage $x = \ell$.
2. Set $C_H = (\ell, p_H \ell)$.
3. Set $C_L = (x_L, p_L x_L)$.
4. Solve $U_H(C_H) = U_H(C_L)$.
5. Check low IC and IR.
6. Test deviations attracting low types only and both types.

17.3 Template C: Credit with collateral

Pair with Visual Intuition Curriculum: Draw high-risk no-collateral contract and low-risk collateral contract; high risk is indifferent, low risk strictly prefers collateral contract.

Reusable template

1. Write borrower utility $U_q(r, x)$ and lender profit $\Pi_q(r, x)$.
2. Complete-information benchmark: zero collateral and zero-profit interest.
3. Candidate: high-risk contract has no collateral; low-risk contract lies on low-risk zero-profit line.
4. High-risk IC binds; solve for x_L .
5. Verify $x_L \in [0, 1]$, low IC, IR, and no entry.

17.4 Template D: Full-insurance continuum

Pair with Visual Intuition Curriculum: Draw cutoff type and average cost of the selected pool.

Reusable template

1. Define $\tau(\theta)$ from insured utility equals uninsured utility.
2. Buyers at premium t are $\{\theta : \tau(\theta) \geq t\}$.
3. Zero profit: $t = \ell E[\theta \mid \tau(\theta) \geq t]$.
4. Solve cutoff fixed point and include no-trade if no profitable entry.

18 Graphical tools

Pair with Visual Intuition Curriculum: Every graph is an equation with slopes labeled; unlabeled curves do not earn proof credit.

18.1 Insurance graph

Pair with Visual Intuition Curriculum: Use state-contingent wealth: accident wealth on x -axis, no-accident wealth on y -axis, and full insurance on the 45-degree line.

In state wealth coordinates (w_a, w_n) , full insurance is $w_a = w_n$. A zero-profit line for risk p has slope determined by actuarial fairness. High-risk indifference curves are steeper in coverage-premium space because high-risk consumers value loss-state wealth more. Label: high-risk full insurance, low-risk partial insurance, and the high-risk indifference curve through the high-risk contract.

18.2 Credit graph

Pair with Visual Intuition Curriculum: Plot r vertically and x horizontally; zero-profit lines slope downward because collateral substitutes for interest from the lender's perspective.

Borrower indifference curves slope downward:

$$\left. \frac{dr}{dx} \right|_{U_q} = -\frac{q}{1-q}.$$

Lender zero-profit lines under the benchmark expression slope downward:

$$\left. \frac{dr}{dx} \right|_{\Pi_q=0} = -\frac{qb}{1-q}.$$

Because $b < 1$, collateral is a costly screening instrument. High-risk borrowers have steeper indifference curves and are less willing to post collateral for a lower interest rate.

18.3 Menu lattice graph

Pair with Visual Intuition Curriculum: Plot type on the horizontal axis and assigned quality on the vertical axis; monotonicity rules out downward jumps.

A finite menu is a staircase. Single crossing rules out a downward step. Adjacent IC constraints pin down transfers, and increasing differences makes local IC imply global IC.

18.4 Lemons graph

Pair with Visual Intuition Curriculum: Plot $B(p)$ against p ; the 45-degree line separates buyer acceptance from rejection.

The conditional expectation curve $B(p)$ must be recomputed after any legal cutoff or participation change. If $B(p) < p$ at a proposed price, buyers reject; if sellers with $x > p$ were included in your expectation, the calculation is inconsistent.

19 Closed-form catalog

Pair with Visual Intuition Curriculum: Each formula has a graph: zero-profit line, no-mimic line, or cutoff fixed point.

Object		Formula	When valid
Two-type transfers	quasilinear	$T_L = v_L(q_L), T_H = v_H(q_H) - v_H(q_L) + v_L(q_L)$	Low IR and high IC bind
High-type quantity		$v'_H(q_H) = c'(q_H)$	Interior top type; no upper bound
Low-type quantity		$(1 - \beta)(v'_L - c') = \beta(v'_H - v'_L)$ at q_L	Interior low type
Insurance slope	indifference	$dt/dx = \frac{pu'(w-\ell+x-t)}{pu'(w-\ell+x-t) + (1-p)u'(w-t)}$	Premium paid in both states
High-risk full insurance		$x_H = \ell$	Monopoly insurance top risk; no upper/lower corner blocking
RS low contract		$U_H(\ell, p_H \ell) = U_H(x_L, p_L x_L)$	Separating candidate
Full-insurance WTP		$u(w - \tau(\theta)) = \theta u(w - \ell) + (1 - \theta)u(w)$	Single full-insurance contract
Log WTP		$\tau(\theta) = 1 - (1 - \ell/w)^\theta w$ when $w = 1$ form gives $1 - 0.2^\theta$ for $\ell = .8$	Log utility; normalized example
Credit high-risk contract		$(r_H, x_H) = (q_H/(1 - q_H), 0)$	Complete info or separating candidate

Object	Formula	When valid
Credit low collateral	$x_L = \frac{q_H - q_L}{q_H(1 - q_L) - b q_L(1 - q_H)}$	Candidate under benchmark profit expression; if $x_L \in [0, 1]$
Lemons buyer value	$B(p) = \frac{\int_x^p v(z)f(z)dz}{\int_x^p f(z)dz}$	Sellers with $x \leq p$ sell

20 Exam variations and how the archetype mutates

Pair with Visual Intuition Curriculum: Draw the same IC/IR skeleton under each variation; only the contract labels and profit equations change.

- **2013 August Q1, monopoly insurance with loading:** known risk type, not screening. Focus on participation binding, less-than-full insurance under loading, and comparative statics with risk aversion.
- **2014 June Q3, insurance menu:** two private risk types. Focus on writing the full constrained maximization, proving SSCP/monotonicity, and proving high-risk full coverage.
- **2019 June Q4, credit screening:** competitive firms and collateral. Focus on zero-profit lines, separating candidate, feasibility of collateral, and no-deviation/existence.
- **2020 June Q2, competitive insurance:** full-insurance-only continuum first, then two-type screening extension. Focus on defining equilibrium, proving existence, finding no-trade/trade fixed points, and drawing two-type separating contracts.
- **2021 June Q3, warranty screening:** product warranty and private valuation. Focus on revelation principle, CARA certainty equivalent, supermodularity, binding low PC/high IC, and top no distortion.
- **2023 June Q4, finite-type quasilinear screening:** pure proof problem. Focus on SSCP iff increasing differences, monotone selection, and adjacent binding.

21 Common traps

Pair with Visual Intuition Curriculum: Draw a red mark next to every constraint you did not use; those are the constraints you must verify.

1. Writing only the IC constraints you expect to bind. Always write all IC and IR first.
2. Confusing signaling and screening. In screening, the uninformed side offers the menu; in signaling, the informed side chooses the signal.
3. Treating a separating candidate in a competitive market as an equilibrium without checking deviations.
4. Pricing insurance by the unconditional average loss probability after only high-risk consumers buy.
5. Forgetting the outside option $(0, 0)$ in monopoly menus.
6. Reversing the order of types in insurance: high risk values coverage more, low risk is less willing to pay for coverage.
7. Reversing the order in credit: high default risk dislikes collateral more because default is more likely.

8. Forgetting that collateral is inefficient when $b < 1$; it screens but destroys surplus.
9. Claiming both types get full insurance under private information.
10. Using local IC without proving global IC or monotonicity.
11. Ignoring bounds such as $x \in [0, \ell]$ or collateral $x \in [0, 1]$.
12. Confusing premiums/transfers: in insurance t is paid in both states, while x is paid only in the loss state.
13. Forgetting that higher transfer can be worse for the agent when transfer is a payment to the principal.
14. Saying “risk aversion increases insurance demand” without tying it to the exact comparative-static object.
15. Hiding proof in a picture. Use the picture to guide inequalities, then write the inequalities.

22 Exercise bank

Pair with Visual Intuition Curriculum: Work these in four passes: setup-only, proof-only, computation-only, and final verification-only.

22.1 Mechanical drills (30 percent)

Pair with Visual Intuition Curriculum: Mechanical drills are the algebraic warm-up: no economic interpretation until every constraint is correctly written.

- M1. **Skill: IC/IR setup.** For two types $\theta_L < \theta_H$ and contracts (q_i, T_i) with utility $f(q, \theta) - T$, write all IC and IR constraints with outside option zero.
- M2. **Skill: insurance utility.** Write $U_i(x, t)$ for loss probability p_i and derive dt/dx along an indifference curve.
- M3. **Skill: zero-profit lines.** In insurance with profit $t - p_i x$, draw and write the zero-profit line for type i .
- M4. **Skill: credit zero-profit.** Using $\Pi_q = (1 - q)r - q + qbx$, solve the zero-profit interest rate r as a function of collateral x .
- M5. **Skill: full-insurance cutoff.** Define the willingness-to-pay function $\tau(\theta)$ for full insurance.
- M6. **Skill: lemons expectation.** If $x \sim U[0, 1]$ and buyer value is αx , compute $B(p) = E[\alpha x \mid x \leq p]$.

22.2 Proof and characterization drills (30 percent)

Pair with Visual Intuition Curriculum: Each proof drill should fit on one page: assumptions, two inequalities, conclusion, verification.

- P1. **Skill: monotone menus.** Prove that IC plus strict increasing differences implies θ_H receives weakly higher q than θ_L .
- P2. **Skill: binding constraints.** In the two-type quasilinear monopoly problem, prove that low IR binds at the optimum under the standard assumptions.
- P3. **Skill: top no distortion.** Prove that the high-risk type receives full insurance in monopoly insurance screening.

- P4. **Skill: RS candidate.** Derive the Rothschild–Stiglitz separating candidate and state the no-deviation test.
- P5. **Skill: global IC from adjacent IC.** For finite types with monotone quantities and increasing differences, show adjacent downward IC implies global downward IC.
- P6. **Skill: CARA warranty.** Derive $v(\theta, w)$ and prove $v_{\theta w} > 0$ or the equivalent increasing-differences property.

22.3 Past-qual applications (25 percent)

Pair with Visual Intuition Curriculum: For each old-qual application, write the first ten lines exactly as you would on the exam.

- Q1. **2014 June Q3 style.** Set up the monopoly insurance screening problem with two risk types and prove the menu is monotone.
- Q2. **2019 June Q4 style.** Derive the credit screening candidate and identify the parameter condition for collateral feasibility.
- Q3. **2020 June Q2 style.** For $u(c) = \log c$, $w = 1$, $\ell = 0.8$, solve the full-insurance cutoff equation numerically to one decimal place.
- Q4. **2021 June Q3 style.** Set up the warranty screening menu using the revelation principle and identify which constraints bind.
- Q5. **2023 June Q4 style.** Prove that $u(q, T, \theta) = f(q, \theta) - T$ has SSCP in $((q, T), \theta)$ iff f has strictly increasing differences in (q, θ) .

22.4 Trap / economist-claim questions (15 percent)

Pair with Visual Intuition Curriculum: Trap questions should be answered by a counterexample, a missing constraint, or a no-deviation check.

- T1. **Claim:** “In any competitive insurance equilibrium, firms earn zero profit in expectation across all potential consumers.” True or false?
- T2. **Claim:** “If a separating menu satisfies both types’ IC constraints, it is automatically a competitive equilibrium.” True or false?
- T3. **Claim:** “Collateral improves efficiency in the credit screening model because it lowers the interest rate.” True or false?

23 Answer sketches

Pair with Visual Intuition Curriculum: The answer sketch format is setup, key equation, result, intuition, mistake to avoid.

23.1 Mechanical sketches

- M1. IC_L: $f(q_L, \theta_L) - T_L \geq f(q_H, \theta_L) - T_H$; IC_H: $f(q_H, \theta_H) - T_H \geq f(q_L, \theta_H) - T_L$; IR_i: $f(q_i, \theta_i) - T_i \geq 0$. Mistake: writing only two constraints.
- M2. $U_i = p_i u(w - \ell + x - t) + (1 - p_i)u(w - t)$. Set $dU = 0$ to get $dt/dx = \frac{p_i u'(w - \ell + x - t)}{p_i u'(w - \ell + x - t) + (1 - p_i)u'(w - t)}$. Mistake: forgetting premium reduces wealth in both states.

- M3. Zero profit is $t - p_i x = 0$, so $t = p_i x$. In (x, t) space the slope is p_i ; high risk line is steeper. Mistake: using average p in a separating contract.
- M4. $(1 - q)r - q + qbx = 0$ gives $r = q(1 - bx)/(1 - q)$. Mistake: forgetting the default loss term needed to match the complete-information benchmark.
- M5. $\tau(\theta)$ solves $u(w - \tau) = \theta u(w - \ell) + (1 - \theta)u(w)$. Buyers at t have $\tau(\theta) \geq t$. Mistake: treating all consumers as buyers.
- M6. $B(p) = \alpha E[x \mid x \leq p] = \alpha p/2$ for $p \in [0, 1]$. Trade requires $\alpha p/2 \geq p$, so for $p > 0$ requires $\alpha \geq 2$. Mistake: using $E[x] = 1/2$ unconditionally.

23.2 Proof/characterization sketches

- P1. Add the two IC inequalities. If $q_L > q_H$, strict increasing differences gives the opposite strict inequality. Hence $q_H \geq q_L$. Mistake: assuming monotonicity before proving it.
- P2. If low IR slack, raise both transfers slightly or at least raise the low transfer while preserving IC until a constraint binds; under standard ordering the first binding lower constraint is low IR. In the reduced solution, $T_L = v_L(q_L)$. Mistake: raising a transfer without checking IC direction.
- P3. Along high-risk IC, $dt/dx > p_H$ when $x < \ell$, so increasing coverage and premium keeps high utility fixed and raises profit. Continue until $x_H = \ell$. Mistake: saying full coverage follows from risk aversion alone.
- P4. Candidate: $C_H = (\ell, p_H \ell)$ and $C_L = (x_L, p_L x_L)$ with $U_H(C_H) = U_H(C_L)$. Then test profitable deviations attracting low only or both. Mistake: omitting the deviation test.
- P5. For $i > j$, sum adjacent IC inequalities or use increasing differences to show each higher type prefers the higher rung. Monotonicity makes local constraints imply global ones. Mistake: proving only downward but not upward IC if both are needed.
- P6. CARA gives $v(\theta, w) = -r^{-1} \log(qe^{-r\theta} + (1 - q)e^{-rw})$. Since $v_w = (1 - q)e^{-rw}/[qe^{-r\theta} + (1 - q)e^{-rw}]$ increases in θ , v has increasing differences. Mistake: differentiating expected utility instead of the certainty equivalent.

23.3 Past-qual sketches

- Q1. Write $U_i(x, t)$, all IC/IR, and objective $\lambda[t_H - p_H x_H] + (1 - \lambda)[t_L - p_L x_L]$. Monotonicity follows from high risk valuing coverage more. Use high full coverage and low IR/high IC binding. Mistake: assuming both full coverage.
- Q2. Use $r_H = q_H/(1 - q_H)$, $x_H = 0$, low zero profit, and high IC. Solve $x_L = \frac{q_H - q_L}{q_H(1 - q_L) - q_L(1 - q_H)}$ and require $x_L \leq 1$. Mistake: not checking $x_L \in [0, 1]$.
- Q3. Solve $1 - 0.2^{\theta^*} = 0.4(1 + \theta^*)$. The interior root is about 0.739, so to one decimal 0.7 and $t \approx 0.696$. Mistake: counting the endpoint zero-mass root as a trade equilibrium.
- Q4. Direct menu: $(p_L, w_L), (p_H, w_H)$. CARA reduces utility to $v(\theta, w) - p$; v is supermodular. Binding constraints are low IR and high IC. Mistake: omitting the outside option $u(0)$.
- Q5. Compare $u(q', T', \theta') - u(q, T, \theta')$ and $u(q', T', \theta) - u(q, T, \theta)$. Transfers cancel, so SSCP is exactly strict increasing differences of f . Mistake: letting the transfer difference obscure the cancellation.

23.4 Trap sketches

- T1. False as stated. Zero profit applies to the pool that actually buys a given active contract, not all potential consumers. Adverse selection changes the buyer pool. Mistake: unconditional averaging.
- T2. False. Competitive equilibrium also requires zero profit and no profitable entrant deviation. A self-selecting menu can be destroyed by cream-skimming. Mistake: using monopoly feasibility as competitive equilibrium.
- T3. False. Collateral can lower interest but is inefficient because the borrower loses x while the lender realizes only $bx < x$ in default. It screens types at a surplus cost. Mistake: equating lower interest with higher total surplus.

24 Past-qual practice map

Pair with Visual Intuition Curriculum: Mark each old qual by the first equation you should write; the equation is the recognition cue.

Year / question	Archetype		First equations to write	Main verification
August 2013 Q1	Monopoly with known loading	insurance type and	$\max t - \gamma p x$ s.t. $U(x, t) \geq U(0, 0)$	Underinsurance; risk aversion comparative statics; no-trade threshold
June 2014 Q3	Monopoly screening	insurance	$U_i(x, t)$, IC/IR, profit $t - p_i x$	SSCP, monotonicity, high-risk full coverage
June 2019 Q4	Competitive screening	credit	$U_q(r, x)$, $\Pi_q(r, x)$, zero-profit lines	Unique candidate; feasibility; no deviation
June 2020 Q2	Competitive full insurance / two-type screening extension	full insurance	$\tau(\theta)$ and $t = \ell E[\theta \mid \text{buyers}]$	Existence; no trade; trade fixed point; two-type diagram
June 2021 Q3	Warranty screening		CARA $v(\theta, w) - p$; IC/IR menu	Supermodularity; binding low PC/high IC; solve menu
June 2023 Q4	Finite-type screening	quasilinear	IC inequalities and increasing differences	Monotone selection; adjacent indifference
August 2023 Q3	Multi-type bridge	signaling	No-mimic recursion for education	Separating recursion; zero-profit wages
August 2024 Q3	Threshold bridge	signaling	Indifference curves under education/wage rules	Pooling/separating threshold verification

Year / ques- tion	Archetype	First equations to write	Main verification
Spring 2015 Q3	Lemons with legal cut-off	$B(p, r) = E[\alpha x \mid r \leq x \leq p]$	Market breakdown and minimum legal standard

25 Final checklist before old quals

Pair with Visual Intuition Curriculum: Use the checklist as a final scan of your graph and equations before turning the page.

1. Did I state the private type, prior probabilities, and who observes what?
2. Did I define the contract space and outside option?
3. Did I write $U_i(c_j)$ clearly enough to form every IC constraint?
4. Did I write all IC and IR constraints before reducing?
5. Did I prove or invoke single crossing before using monotonicity?
6. In monopoly screening, did I justify which constraints bind?
7. In competitive screening, did I impose zero profit only on active contracts and then test deviations?
8. Did I check all bounds: coverage, collateral, nonnegative quantity, nonnegative consumption?
9. Did I label every graph slope: indifference, zero profit, full insurance, pooled cost?
10. Did I distinguish separating, pooling, partial pooling, no trade, and market breakdown?
11. Did I verify dropped constraints and not just solve the binding equations?
12. Did I include one sentence of economic intuition after the algebra?

26 Quality audit

Pair with Visual Intuition Curriculum: The quality audit is the same as the exam audit: every conclusion must be backed by a constraint or deviation check.

This packet inventories the accessible sources; identifies the module's past-qual appearances; states all canonical models with primitives, timing, choices, constraints, equilibrium/optimal conditions, and verification; includes source-specific exam variations; gives derivation templates and a closed-form catalog; and includes an exercise bank with answer sketches. Remaining limitation: the 2013–2017 scanned exams have limited machine-readable text, so their entries are based on rendered page inspection rather than full OCR.